^{3}He$(n,\cdot)^{4}$He capture cross sections and the IPC pair yield

The measured M1+E1 radiative capture, the γ-dark E0, and the e^+e^- yield that follows from $\sigma\times$(pair fraction)

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Internal note — informal. Drafted by Claude (Anthropic AI assistant) under D. Neff's direction; physics, numbers and text reviewed by D. Neff.

Part 0 — Summary

In plain words. The E0 transition is interesting but it does *not* rescue the thermal measurement. As the neutron energy drops, the γ-dark E0 mode does turn on — but its capture cross section is still expected to be ∼100–1000× smaller than the M1 ($\sigma_{\rm E0} = f\,\sigma_{\rm M1}$, $f \sim 10^{-2}$–10^{-3}). That deficit is offset because E0 is γ-forbidden, so *every* E0 capture makes an e^+e^- pair, whereas only $\sim 3.5\times10^{-3}$ of M1 photons internally convert — E0 *skips* that penalty (a ∼300× relative boost). The two $\sim 10^{-3}$ factors roughly cancel, leaving E0 pair (hence X17) production *the same order as* the M1 in the sub-keV: at best a factor of ∼2–3 on the thermal statistics, **not nearly enough to save that measurement**. At high E_n the E0 adds essentially nothing, since the M1+E1 (E1/giant-dipole) dominates and the E0 has died away. *This is outside the author's expertise and pushes the current limit of what AI can do; the calculations were cross-checked but the note should be read skeptically — insights and holes welcome.*

The capture cross section factorises as $\sigma = \sigma_{\rm form}\times(\Gamma_X/\Gamma_{\rm tot})$ — **form the ^{4}He* state, then radiate to the ground state rather than break up.** Two consequences drive this note:

- **The photon-emitting capture (M1+E1) is MEASURED** — it is the ENDF ^{3}He$(n,\gamma)^4$He cross section (55 µb at thermal, the classic value), with the M1-dominated $1/v$ at low E_n and the E1/GDR rise toward MeV. **Only the γ-dark E0 is estimated:** $\sigma_{\rm E0} = f\,\sigma_{\rm M1}$, $f\sim10^{-3}$–10^{-2}, sitting 2–3 decades below at thermal (and further below toward MeV, as the E1 rises and the E0 dies).
- **The IPC (e^+e^-) yield is just $\sigma\times$(pair fraction):** $\times\alpha_{\rm IPC}\approx3.5\times10^{-3}$ for M1+E1 (only the internal-pair tail of the photon makes a pair), $\times1$ for E0 (γ-dark $\Rightarrow$ 100 % pairs). Multiplying the measured curve *down* by $\alpha_{\rm IPC}$ while leaving E0 alone lifts E0 by $1/\alpha_{\rm IPC}\approx285$: **the two pair yields become COMPARABLE at thermal/sub-keV** (E0/M1E1 $= f/\alpha_{\rm IPC}\approx$ 0.3–3), while M1+E1 wins at MeV.
- **The dominant uncertainty is f** (∼1 decade); the measured $\sigma_{n\gamma}$ is firm and $\alpha_{\rm IPC}$ contributes only $\sim\times1.8$.

1 The cross section is a product: form × radiate

For capture through a compound ^{4}He* state of spin J,

$$\sigma(n,X) \;=\; \underbrace{\frac{\pi}{k^2}\,g_J\,\frac{\Gamma_n(E)\,\Gamma_{\rm tot}}{(E-E_r)^2+(\Gamma_{\rm tot}/2)^2}}_{\sigma_{\rm form}\quad\text{(form the state)}} \times \underbrace{\frac{\Gamma_X}{\Gamma_{\rm tot}}}_{\text{(radiate as }X)} .$$

The total width is almost all **breakup** ($\Gamma_{\rm tot} \approx \Gamma_n + \Gamma_p$, the (n,p)t channel), so the radiative branch is a $\sim 10^{-8}$ sliver — that is why capture is rare. Per channel, $\sigma_{\rm M1} = \sigma_{\rm form}(1^+)\,\Gamma_{\rm M1}/\Gamma_{\rm tot}$ and $\sigma_{\rm E0} = \sigma_{\rm form}(0^+)\,\Gamma_{\rm E0}/\Gamma_{\rm tot}$, so

$$f \equiv \frac{\sigma_{\rm E0}}{\sigma_{\rm M1}} = \underbrace{\frac{\sigma_{\rm form}(0^+)}{\sigma_{\rm form}(1^+)}}_{\sim 1\text{–}2 \text{ (rough)}} \times \underbrace{\frac{\Gamma_{\rm E0}}{\Gamma_{\rm M1}}}_{\text{the unknown}} .$$

(^{4}He capture is mostly *direct*, so "form then decay" is not a literal two-step process — but the cross section still separates into a kinematic/penetrability factor × an EM-strength factor, which is what matters here. The formation side is detailed in `docs/e0_branch/doorway_states_note.md`.)

1.1 Which ^{4}He* state forms (the formation factor)

The states available near the $n+{}^3$He threshold (20.58 MeV) are the broad, overlapping ^{4}He levels of the TUNL $A = 4$ evaluation [1] (Fig. 1). Two features set the formation:

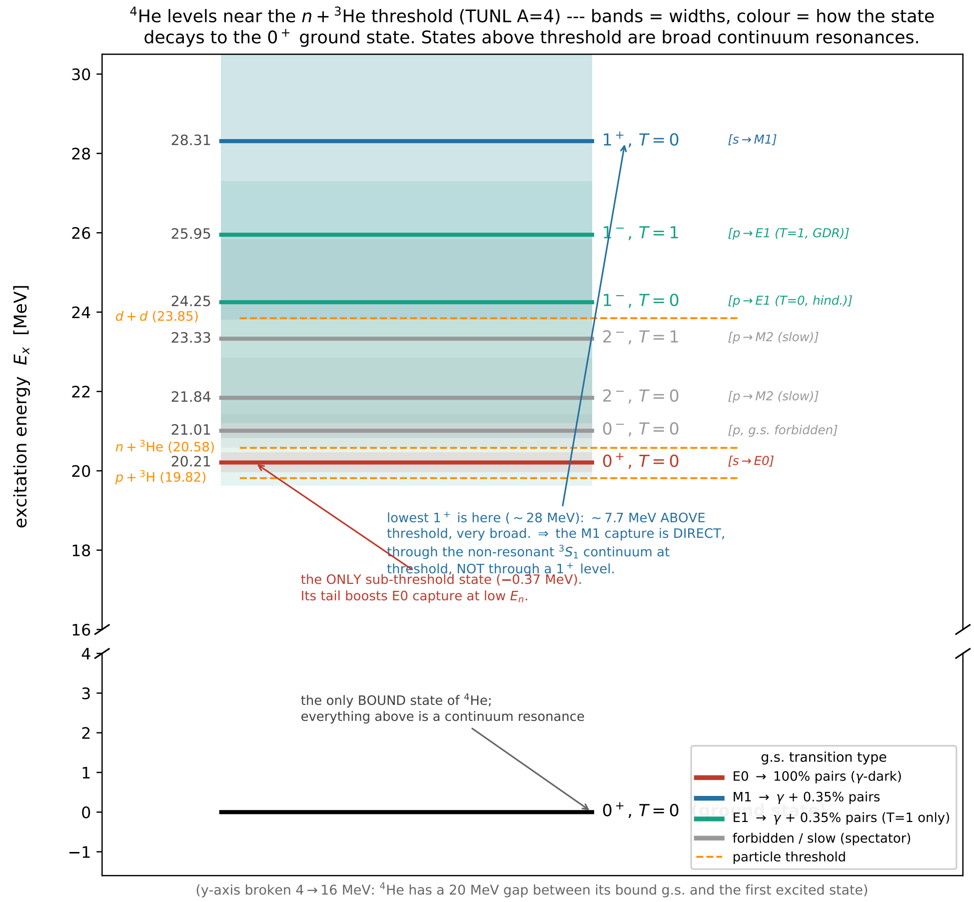


Figure 1: ^{4}He levels near the $n + {}^3$He threshold (TUNL $A = 4$). The levels are broad and overlapping, and the lowest 1^+ sits $\sim$8 MeV *above* threshold — there is no low-lying 1^+.

- **The levels are broad** ($\Gamma \sim 0.5$–13 MeV, so $\Gamma \gtrsim |E_r|$): ^{4}He has no sharp levels above threshold, so capture is **mostly direct/continuum** capture, modulated by broad structure rather than sharp resonant capture.
- **There is no low-lying 1^+ level** (the lowest is $\sim$28 MeV). So the M1 capture proceeds *directly* through the continuum, while only the 0^+ (a sub-threshold level at 20.21 MeV, 0.37 MeV below threshold) has a broad resonance assisting it.

Parity fixes the entrance orbital angular momentum, $\pi = (-1)^\ell$:

reached by	$\to$ states	energy behaviour
s-wave ($\ell = 0$)	0^+, 1^+	no barrier — rise as $1/v$, dominate at low E_n
p-wave ($\ell = 1$)	0^-, 2^-, 1^-	barrier-suppressed — switch on at sub-MeV/MeV

Two consequences feed the cross sections below: the 1^+ (M1) is *pure direct* continuum capture — giving the clean $1/v$, 55 µb ^{3}He(n, γ) with no thermal resonance — while the 0^+ (E0) is direct *plus* a sub-threshold boost ($\sim$10× at threshold relative to 1 MeV), concentrating the E0 at the lowest E_n. The quantitative doorway decomposition (and the correction of an earlier "1^+ dominates" artifact) is in `docs/e0_branch/doorway_states_note.md`.

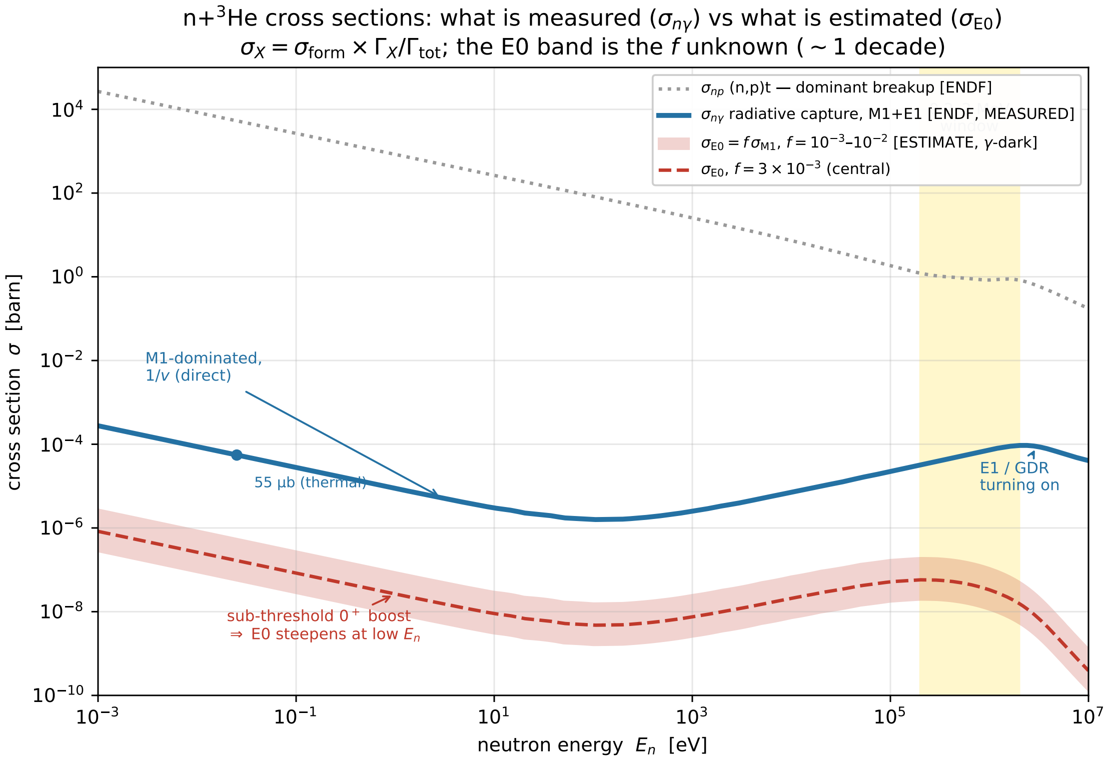
1.2 Which states can radiate to the ground state (the EM-strength factor)

Whatever forms, ^{4}He* almost always just breaks back up ($p + {}^3$H, the (n,p)t reaction) and makes no pair; the rare EM drop to the 0^+ ground state is the only pair source, and *what kind* of drop is fixed by J^π:

State	$\to 0^+$ g.s.	pairs?	role for our signal
0^+ (20.21)	**E0** (γ forbidden)	**100 %**	**E0 pair source** — low E_n
1^+ ($^3S_1$)	M1	0.35 %	**M1 pair source** — the ordinary 55 µb
1^- (24–26)	E1	0.35 %	**E1 pair source** — rises toward MeV
0^- (21.01)	forbidden	—	spectator (breaks up)
2^- (21.84)	M2 (too slow)	$\sim$0	spectator

The decisive line is $0^+ \to 0^+$: a real photon is forbidden ($0 \to 0$ kills every photon multipole, M1 included), so the *only* outlet is an e^+e^- pair — **100 % pairs, γ-dark**, invisible to ENDF/Geant4. The 1^+ and 1^- emit a 20.6 MeV γ with the small internal-pair tail α_{IPC} (so M1 is a *different* capture channel from E0, not its photon partner). The 0^- ($0 \to 0$ plus parity flip forbids even E0) and 2^- (M2 far too slow at 20 MeV) are spectators. This is why the EM-strength factor reduces to the three source states $\{0^+, 1^+, 1^-\}$ used throughout.

2 The capture cross sections: measured vs estimated



Reading the figure:

- σ_{np} **(grey, dotted)** — the dominant (n,p)t breakup, 5330 b at thermal. This *is* Γ_{tot}, and the reference for "rare."
- $\sigma_{n\gamma}$ **(blue, MEASURED)** — total radiative capture (M1+E1), ENDF/B-VIII.0, 55 μb at thermal. The $1/v$ falloff is the M1-dominated direct capture; the rise toward MeV is the E1/giant-dipole strength (the $T = 1\ 1^-$) turning on. **No estimation enters this curve.**
- σ_{E0} **(red band, ESTIMATE)** — the γ-dark E0 capture, $\sigma_{E0} = f\,\sigma_{M1}$ with the sub-threshold 0^+ boost steepening it toward low E_n. The band is $f = 10^{-3}$–10^{-2}; $\sigma_{E0}(\text{thermal}) \approx$ 0.055–0.55 μb. Invisible to $\sigma_{n\gamma}$ (it emits no photon).

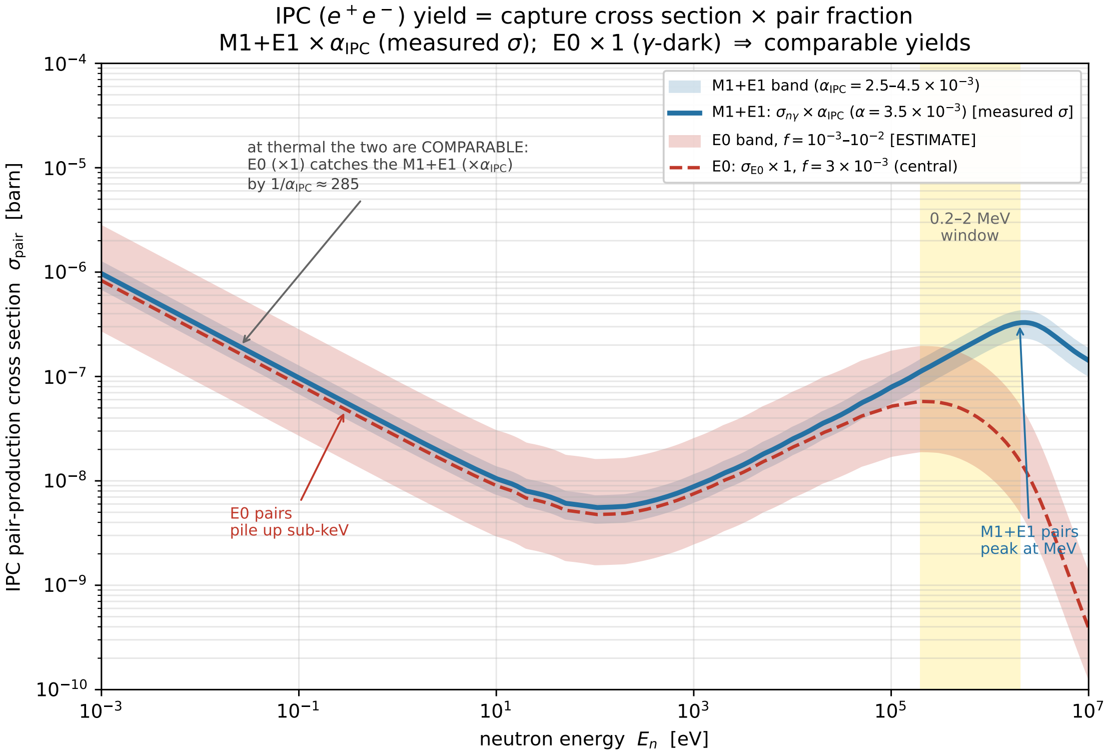
The honest split is stark: a measured photon capture, and a γ-dark E0 capture 2–3 decades below it at thermal (further below toward MeV) whose absolute height is the open f.

3 The IPC pair yield: $\sigma\times$(pair fraction)

The e^+e^- yield is each cross section times its pair fraction — $\times\alpha_{IPC}$ for the photon channel (only the internal-pair tail of the γ becomes a pair), $\times 1$ for E0 (it makes nothing *but* pairs):

$$\sigma_{pair}(\mathrm{M1}+\mathrm{E1}) = \sigma_{n\gamma} \times \alpha_{IPC}, \qquad \sigma_{pair}(\mathrm{E0}) = \sigma_{E0} \times 1.$$

$\alpha_{IPC} \approx 3.5\times10^{-3}$ at 20.6 MeV (anchored to the measured ^{12}C 15.1 MeV M1 IPC, $(3.3\pm0.5)\times10^{-3}$ [6], and the high-energy IPC theory [7]; BrIcc's pair tables stop at 6 MeV [5]; band 2.5–4.5$\times10^{-3}$; Alberto's table uses 2.1×10^{-3}).



The result. Multiplying the measured curve *down* by α_{IPC} while leaving E0 *unchanged* lifts E0 by $1/\alpha_{\text{IPC}} \approx 285$ relative to M1+E1. The 4–5-decade gap of the capture figure **collapses**: the two pair yields are **comparable at thermal** (at $E_n = 25\,\text{meV}$, $\sigma_{\text{pair}}^{\text{M1+E1}} = 0.19\,\mu\text{b}$ vs $\sigma_{\text{pair}}^{\text{E0}} = 0.055$–$0.55\,\mu\text{b}$, i.e. $\text{E0/M1E1} = f/\alpha_{\text{IPC}} \approx 0.3$–$3$). They then split by energy — **E0 pairs pile up sub-keV, M1+E1 pairs peak at MeV** — so the γ-dark channel matters precisely where the ordinary one is dead.

4 X17 production, and the two scenarios

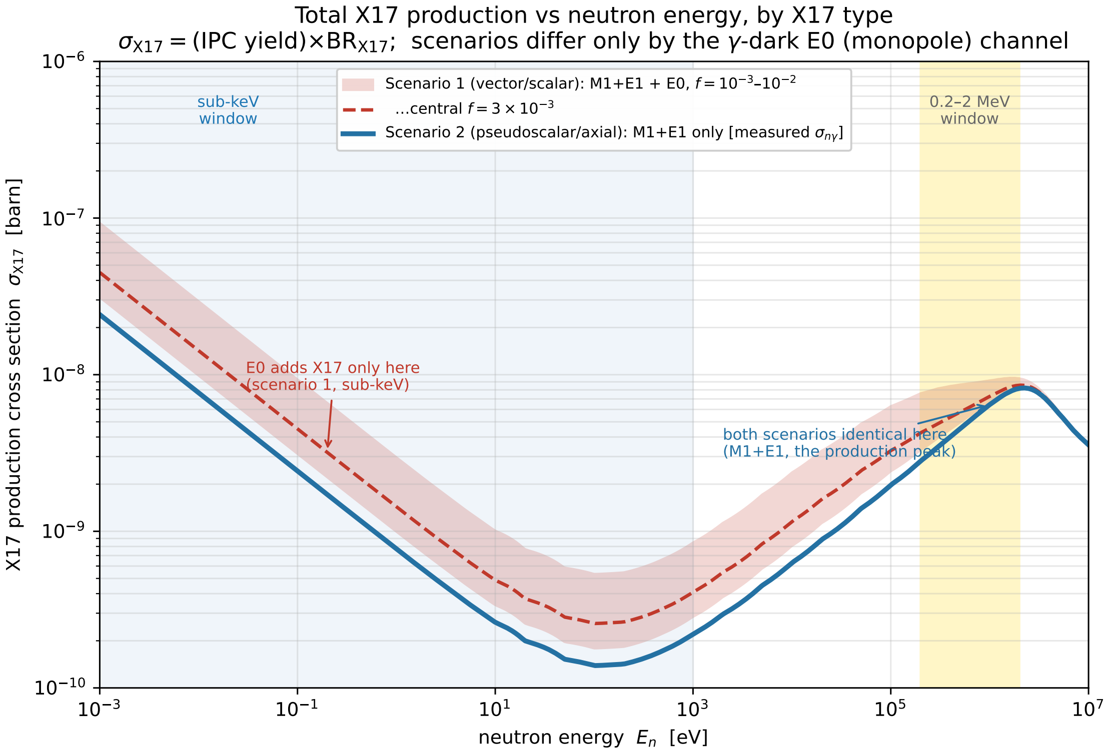
X17 is produced as a sub-branch of the same transitions: $\sigma_{\text{X17}} = (\text{IPC yield}) \times \text{BR}_{\text{X17}}$ with $\text{BR}_{\text{X17}} = 0.025$ (X17 per e^+e^- pair, the project benchmark). Which *modes* contribute depends on the X17 quantum numbers, through one decisive rule: a $0^+ \to 0^+$ (E0/monopole) transition can emit a massive boson only if its J^P allows $0^+ \to 0^+ + X$ —

X17	J^P	$0^+ \to 0^+$ emission
vector (dark photon)	1^-	**allowed** ($L = 1$)
scalar	0^+	**allowed** ($L = 0$)
pseudoscalar	0^-	forbidden
axial-vector	1^+	forbidden

The reason $0^+ \to 0^+$ is γ-dark is specifically that the *photon is massless* (transverse-only, cannot carry a monopole); X17 is massive ($\sim$17 MeV), so that argument does not apply — **E0 is γ-dark but not X17-dark**, and a vector/scalar X17 could even be relatively *enhanced* (it is not competing against the 0.35 % IPC tail of a real γ, the same logic that makes E0 "100 % pairs"). Two honest caveats keep this from being a discovery: (i) the E0→X17 branching is a *separate* unknown — not the 2.5 % measured for the M1 line (different multipole, phase space, coupling); we use 2.5 % only as a placeholder. (ii) Whatever the branching, the rate is bounded by the (small) E0 *capture* rate, so X17-from-E0 stays $\lesssim$0.3/day produced (§6, §5).
so the total X17 production splits into two scenarios, differing *only* by the γ-dark E0 channel:

- **Scenario 1 (vector or scalar):** M1+E1 + E0.
- **Scenario 2 (pseudoscalar or axial):** M1+E1 only.

(At the strict per-multipole level the M1/E1 couplings are also type-dependent; we take "X17 rides the measured M1+E1 strength" as the working approximation, so the clean scenario difference is the E0 piece.)



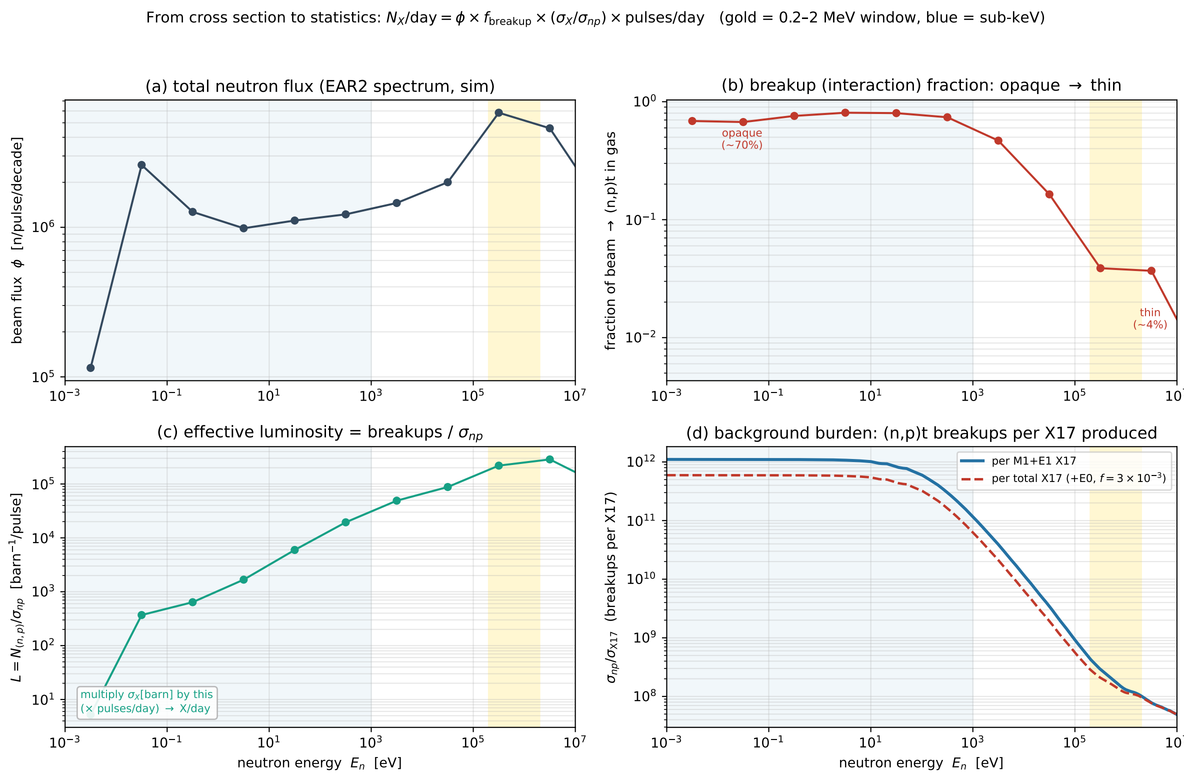
The two scenarios are **identical at the MeV production peak** (all M1+E1) and diverge only in the sub-keV region, where scenario 1 gets the E0 band.

5 From cross section to statistics

To turn a cross section into a rate we fold in the simulated beam:

$$\frac{N_X}{\text{day}}(E_n) = \underbrace{\phi(E_n)}_{\text{flux}} \times \underbrace{f_{\text{breakup}}(E_n)}_{\substack{\text{fraction of beam}\\ \text{interacting in gas}}} \times \underbrace{\frac{\sigma_X(E_n)}{\sigma_{np}(E_n)}}_{\text{X per breakup}} \times \text{pulses/day},$$

equivalently $N_X = N_{(n,p)} \times \sigma_X/\sigma_{np}$ — **the (n,p)t breakup, counted directly in Geant4, is the luminosity monitor.** The clean part: even where the gas is *opaque* (sub-keV), the flux attenuation along the column is set by the dominant (n,p) absorption, so it multiplies *both* channels by the same $(1-e^{-\tau_{np}})$ and **cancels in the ratio** — $N_X = N_{(n,p)}\,\sigma_X/\sigma_{np}$ holds exactly. (For M1+E1 we instead use the direct Geant4 (n,γ) count, which bakes in flux/geometry/opacity already; E0, absent from Geant4, is the one obtained via the ratio.)



The four factors, vs energy: **(a)** the EAR2 flux ϕ; **(b)** the breakup fraction, $\sim$70 % (opaque) sub-keV falling to $\sim$4 % (thin) at MeV; **(c)** the effective luminosity $L = N_{(n,p)}/\sigma_{np}$ — multiply σ_X[barn] by this ($\times$pulses/day) to get X/day; **(d)** the background burden, (n,p)t *breakups per X17 produced*, which falls from $\sim 10^{12}$ in the sub-keV to $\sim 10^8$ at MeV. Panel (d) is the quantitative reason the MeV window is favourable: $\sim 10^4\times$ fewer breakups per signal than sub-keV. (The E0 channel, scenario 1, lowers the burden at low E_n — red dashed — but only into the $\sim 10^{11}$ range.)

The daily statistics (the number we want)

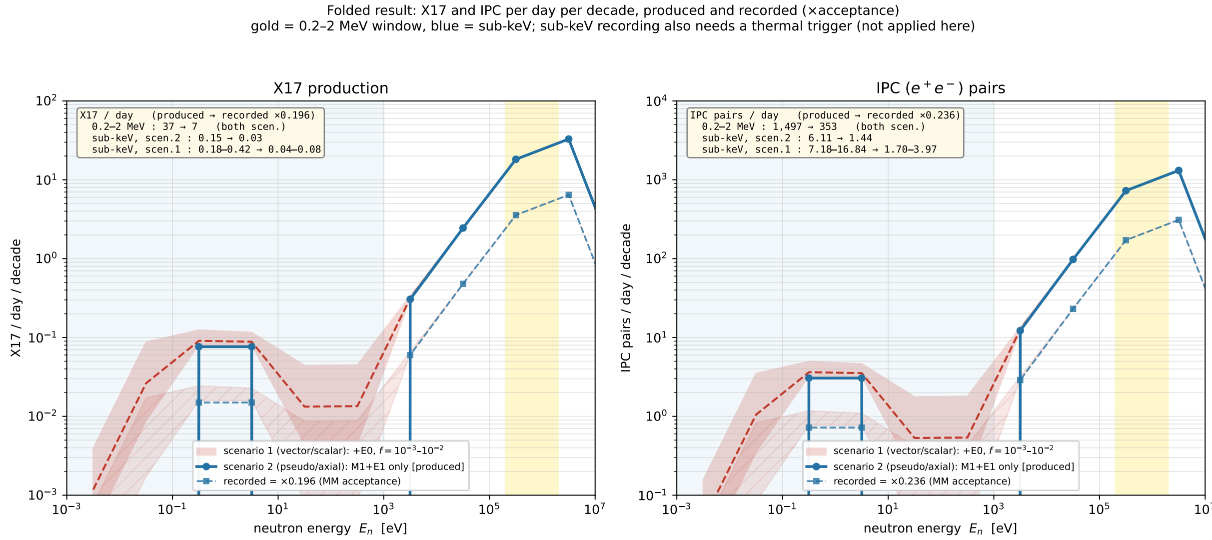
Folding the measured per-decade campaign flux (`mev_rates.json`), the **X17 produced per day** in the two windows:

X17 produced / day	Scenario 2 (M1+E1 only)	Scenario 1 (vector/scalar) M1+E1 + E0, $f = 10^{-3}$–10^{-2}
high-energy window (0.2–2 MeV)	37.4	37.4 (E0 adds < 0.004)
full sub-keV window ($E_n < 1$ keV)	0.15	0.18–0.42

(Anchored $\alpha_{\text{IPC}} = 3.5\times10^{-3}$; the high-energy row is 22.5/day with the table $\alpha_{\text{IPC}} = 2.1\times10^{-3}$ — the committed headline. E0 *alone* contributes 0.03–0.27 X17/day in the sub-keV and < 0.004/day in the high-energy window.)

What this says. (i) The high-energy window carries the X17 rate — $\sim$22–37 produced/day — and is **identical in both scenarios**: E0 is irrelevant there, so the baseline measurement does not depend on the X17 type. (ii) The sub-keV window is **starved either way**: even the most favourable case (vector/scalar X17, $f = 10^{-2}$) gives only $\sim$0.4 X17/day, $\sim 100\times$ below the high-energy window; the E0 channel at best $\sim$2–3$\times$ the (tiny) sub-keV M1+E1 rate. So a vector/scalar X17 would add a *distinct sub-keV signature*, but not significant *statistics*. (iii) These are *produced*; recorded $\approx \times 0.196$ (MM acceptance), and the sub-keV window additionally requires a thermal trigger (the 0.2–2 MeV window is already in the 10 µs flash readout).

Everything folded: per day per decade



The full fold-down, per decade: X17 (left) and IPC pairs (right), *produced* and *recorded* (× the MM-double geometric acceptance, 0.196 X17 / 0.236 IPC). Scenario 2 (blue) is M1+E1 only; scenario 1 (red band) adds E0 over $f = 10^{-3}$–10^{-2}. The MeV peak (gold window) carries the rate and is scenario-independent; the sub-keV (blue) is where scenario 1's E0 lifts an otherwise-dead region — but to small absolute numbers. The boxed sums are the window totals (produced → recorded). *(Acceptance only; the sub-keV recording additionally needs a thermal trigger, and these use at-rest acceptance — the E_n-boost correction is pending.)*

The trigger/readout window: what is actually recorded

One more fold turns "produced" into "recorded": the detector only reads out for a finite time after the proton pulse, and time-of-flight maps that window onto neutron energy, $\mathrm{TOF}(E_n) = L/v$ at $L = 19.5\,\mathrm{m}$. **This is what decides whether E0 is ever seen.**

- The **∼10 µs flash readout** accepts only $E_n \gtrsim 20\,\mathrm{keV}$ (TOF= $10\,\mu\mathrm{s} \Leftrightarrow 19.9\,\mathrm{keV}$). The 0.2–2 MeV window sits comfortably inside it.
- The **E0 pairs are sub-keV** (TOF $> 45\,\mu\mathrm{s}$) — they arrive *after* the flash window closes. **Recorded E0 under the flash readout** ≈ 0. Only a dedicated **thermal trigger** recovers the sub-keV decades.

Folding acceptance (×0.196 X17, ×0.236 IPC) and the two readout schemes onto the windowed produced rates above ($\alpha_{\mathrm{IPC}} = 3.5 \times 10^{-3}$):

recorded / day	produced	flash 10 µs	flash + thermal
X17 signal			
0.2–2 MeV (both scen.)	37.4	7.3	7.3
sub-keV, scen. 2 (M1+E1)	0.15	0	0.03
sub-keV, scen. 1 (+E0)	0.18–0.42	0	0.035–0.083
IPC background pairs			
0.2–2 MeV	1497	353	353
sub-keV, no E0	6	0	1.4
sub-keV, with E0	7–17	0	1.7–4

(With the table $\alpha_{\mathrm{IPC}} = 2.1 \times 10^{-3}$ the 0.2–2 MeV rows scale by 0.6: X17 produced 22.5, recorded 4.4/day — the committed headline. The full-spectrum totals, integrating beyond 0.2–2 MeV, are larger still: ∼32.7 X17/day produced, ∼500 IPC pairs/day recorded under the flash readout — the MeV-note baseline.)

What this means for the measurement.

- **The flash-readout run is unaffected by E0.** Its recorded rate is entirely M1/E1; E0 is sub-keV and arrives after the window, so leaving E0 out of the generator does not bias the baseline high-energy analysis.
- **A thermal trigger is the only way to reach E0**, and the prize is modest: $\lesssim$3 extra IPC pairs/day and $\lesssim$0.06 X17/day, sitting on the enormous sub-keV charged-particle load (∼3×10^5 (n,p)t per pulse in the opaque gas). The statistics exist in thermals, but the E0 pair signal is $f \times 10^{-8}$ of them — **S/B, not rate, is the obstacle.**
- **The generator action item is small and low-priority:** a third pair sub-generator at the 0^+ transition energy with monopole kinematics, weighted by $N_{\mathrm{E0}}/N_{\mathrm{M1/E1}}$; it will not move the recorded spectra under the flash readout.

6 Uncertainties

piece	size of uncertainty	how to shrink
$\sigma_{n\gamma}$ (M1+E1 capture)	small (evaluated data)	— (measured)
α_{IPC} (pair fraction, M1+E1)	$\sim\times1.8$ (2.5–4.5$\times10^{-3}$)	high-energy IPC theory + ^{12}C anchor
$\sigma_{\mathrm{form}}(0^+)/\sigma_{\mathrm{form}}(1^+)$	$\sim\times2$–3	^{4}He R-matrix reduced widths
$f = \sigma_{\mathrm{E0}}/\sigma_{\mathrm{M1}}$	**∼1 decade (10^{-3}–10^{-2})**	**^{4}He R-matrix + (e, e') monopole norm.**

f **dominates by far**, so the E0 pair yield should be quoted as an order-of-magnitude *band*, not a central value (as drawn). The formation ratio and α_{IPC} are sub-dominant and, for f, are already bundled into it. A literature dig found the pieces but not the number: the ^{4}He 0_2^+ E0 strength *is* measured (the MAMI monopole form factor [2], the "α-particle monopole puzzle" [3]) and is a real, collective transition — but that is the *decay-side* matrix element, and turning it into a capture rate needs reaction theory; the ATOMKI ^{4}He pair work [4] models its pairs as E1/M1 capture, not a clean E0 line; and no calculation of ^{3}He$(n, e^+e^-)^4$He exists. The bracket $f \sim 10^{-3}$–10^{-2} is a transparent order-of-magnitude estimate (statistical $^1S_0$ weight $\frac{1}{4}$ as a floor, E0 having no real-photon mode as a ceiling). The one calculation that turns the red band into a line is a ^{4}He multi-level R-matrix (Hale's evaluated system [1]) for the $^1S_0$ $n + {}^3$He→g.s. E0 capture, normalised by the measured ^{4}He(e, e') monopole strength.

Reproducing this note

Figures: `scripts/make_e0_cross_section_fig.py` (capture cross sections, →`fig_e0_xsec`), `scripts/make_e0_ipc_yield_fig.py` (pair yield, → `fig_e0_ipcyield`), and `scripts/make_e0_x17_production_fig.py` (X17 production + the window daily statistics, →`fig_e0_x17`), and `scripts/make_e0_luminosity_fig.py` (the flux/breakup/luminosity bridge, →`fig_e0_lumi`), and `scripts/make_e0_folded_fig.py` (the folded per-day-per-decade result, →`fig_e0_folded`); all read `data/He3.h5` (ENDF/B-VIII.0) and `analysis/mev/mev_rates.json`. The formation physics behind σ_{form} (level diagram, doorway decomposition, continuum-vs-resonance): `docs/e0_branch/doorway_states_note.md` and `formation_and_decay.md`. This note now subsumes the earlier two-part formation/decay note (`e0_pair_channel_note.tex`) and the recorded-metric note (`e0_final_metric_note.tex`); the recorded table here uses `scripts/make_e0_final_metric.py` (→`analysis/e0/final_metric.json`).

References

[1] D. R. Tilley, H. R. Weller, G. M. Hale, “Energy levels of $A = 4$”, Nucl. Phys. A **541** (1992) 1 (TUNL); G. M. Hale’s ^{4}He R-matrix.

[2] S. Kegel *et al.*, “Measurement of the α-Particle Monopole Transition Form Factor”, Phys. Rev. Lett. **130** (2023) 152502 (MAMI A1).

[3] “Derivation of transition density from the observed $^4\mathrm{He}(e,e')^4\mathrm{He}(0_2^+)$ form factor ...”, arXiv:2306.07268; arXiv:2406.00653.

[4] A. J. Krasznahorkay *et al.*, “New anomaly observed in ^{4}He ...”, Phys. Rev. C **104** (2021) 044003; arXiv:2104.10075.

[5] T. Kibédi *et al.*, BrIcc, Nucl. Instrum. Meth. A **589** (2008) 202; https://github.com/IAEA-NSDDNetwork/BrIcc (pair coefficients tabulated only to 6 MeV).

[6] Measured ^{12}C 15.1 MeV M1 IPC, $\alpha_\pi = (3.3 \pm 0.5) \times 10^{-3}$ (PEPSI; arXiv:nucl-ex/9704007).

[7] M. Schlüter, G. Soff, W. Greiner, Z. Phys. A **286** (1978) 149; At. Data Nucl. Data Tables **24** (1979) 509 (high-energy IPC).

[8] R. Wervelman *et al.*, $^3\mathrm{He}(n,\gamma)^4\mathrm{He}$ M1/E1 content, Nucl. Phys. A **526** (1991) 265.